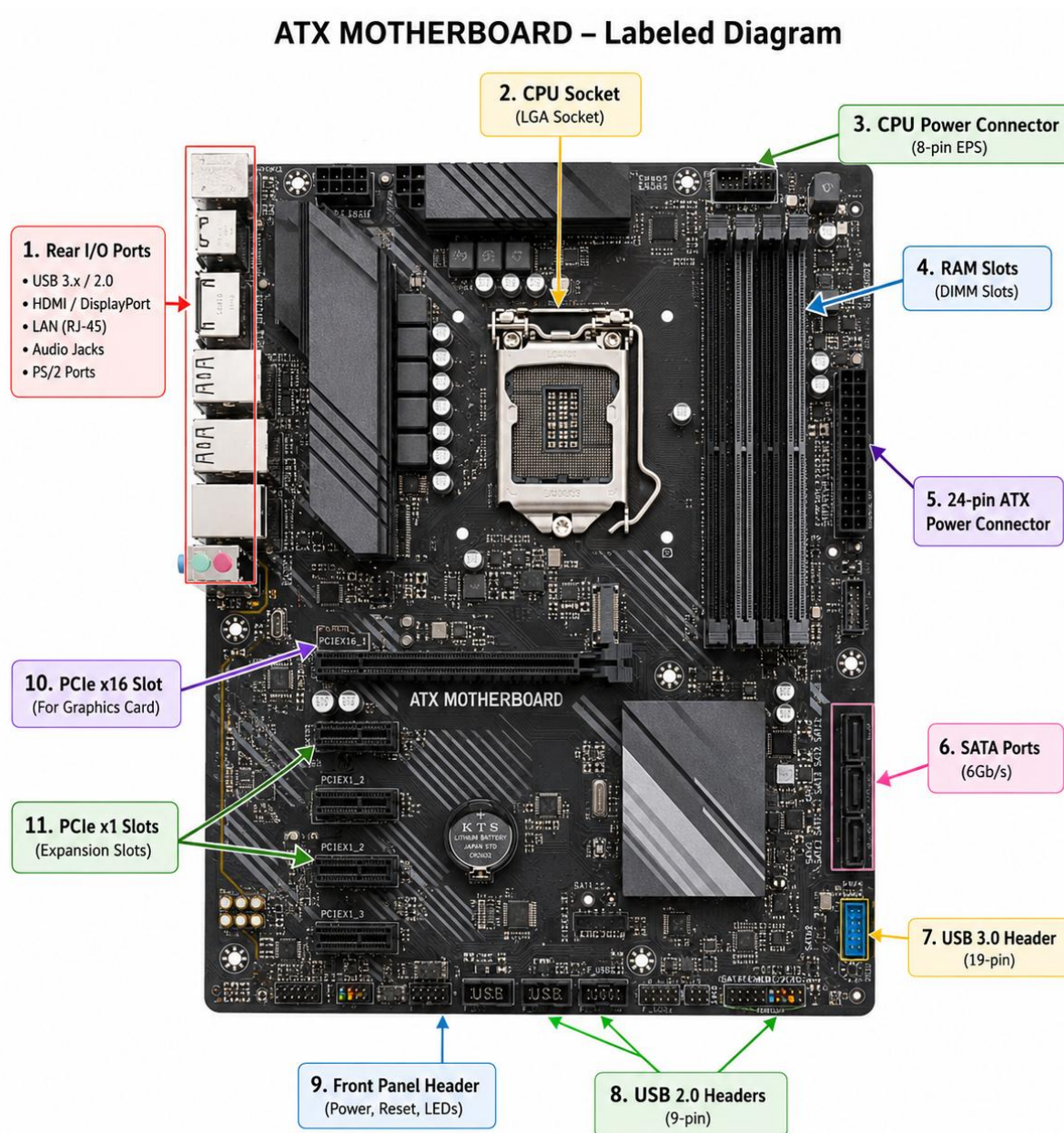


ASSIGNMENT:2

MOTHERBOARD AND CPUTHORY

Q1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).

ANS.



1. **CPU Socket**
2. **RAM Slots (DIMM Slots)**
3. **PCIe x16 Slot**
4. **PCIe x1 Slots**

- 5. SATA Ports
- 6. 24-pin ATX Power Connector
- 7. CPU Power Connector (8-pin EPS)
- 8. USB Headers (USB 2.0/3.0):
- 9. Front Panel Header
- 10. Rear I/O Ports

Q2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

ANS.

CPU Socket Type	Description	One Compatible CPU Model
LGA1151	An Intel CPU socket used for 6th, 7th, 8th, and 9th Generation Intel Core processors (depending on motherboard chipset).	Intel Core i7-9700K
AM4	An AMD CPU socket that supports Ryzen processors and offers excellent compatibility across multiple generations.	AMD Ryzen 5 5600X
LGA1200	An Intel CPU socket designed for 10th and 11th Generation Intel Core processors.	Intel Core i5-10400

Q3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

ANS.

1. CPU (Central Processing Unit):

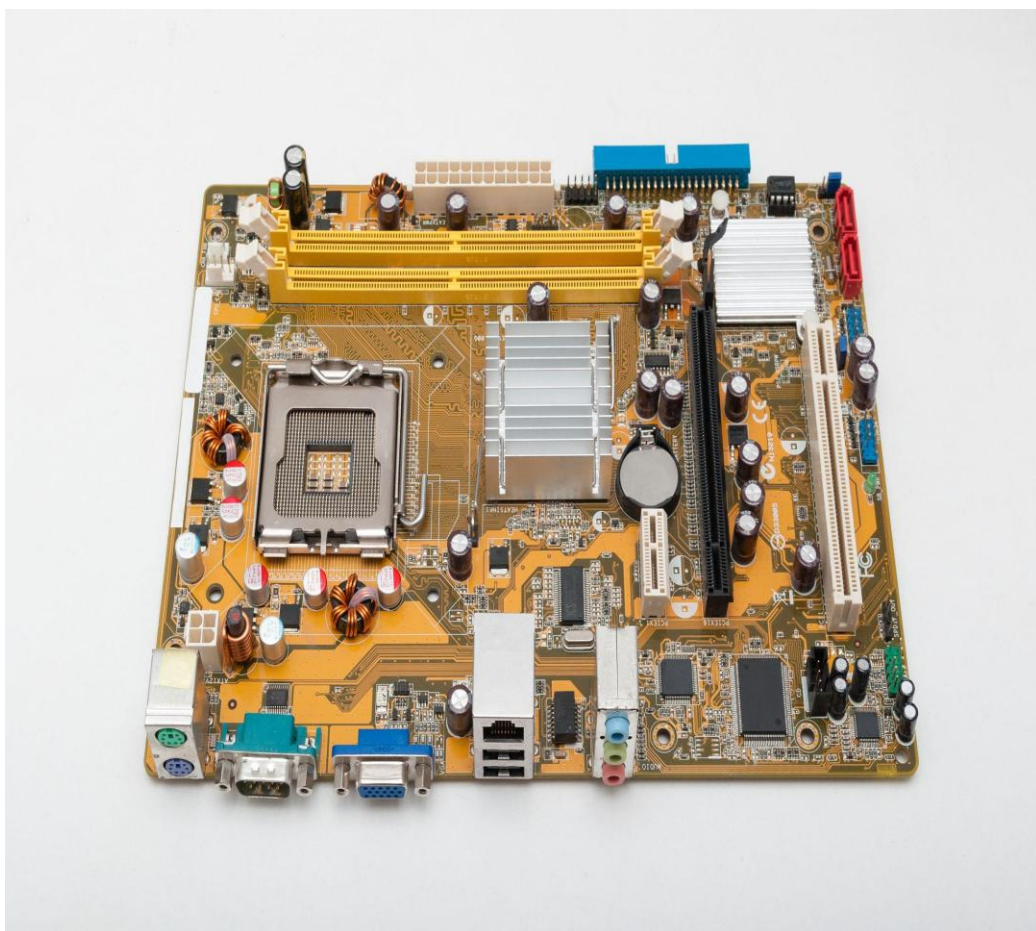
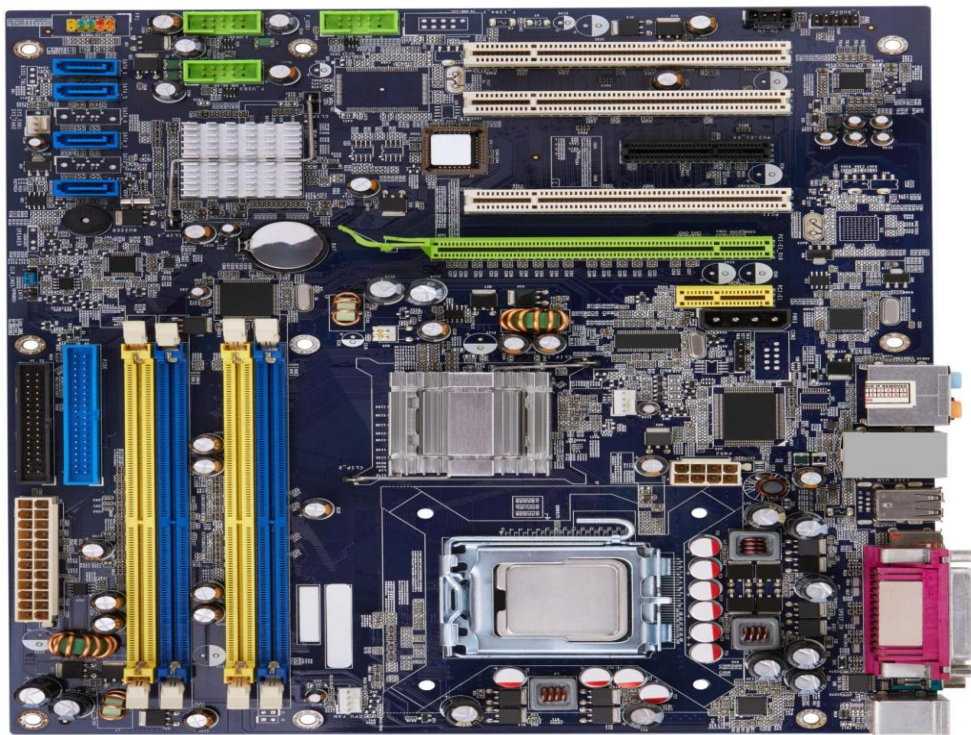
- The CPU is the brain of the computer. It processes instructions, performs calculations, and controls the execution of programs.

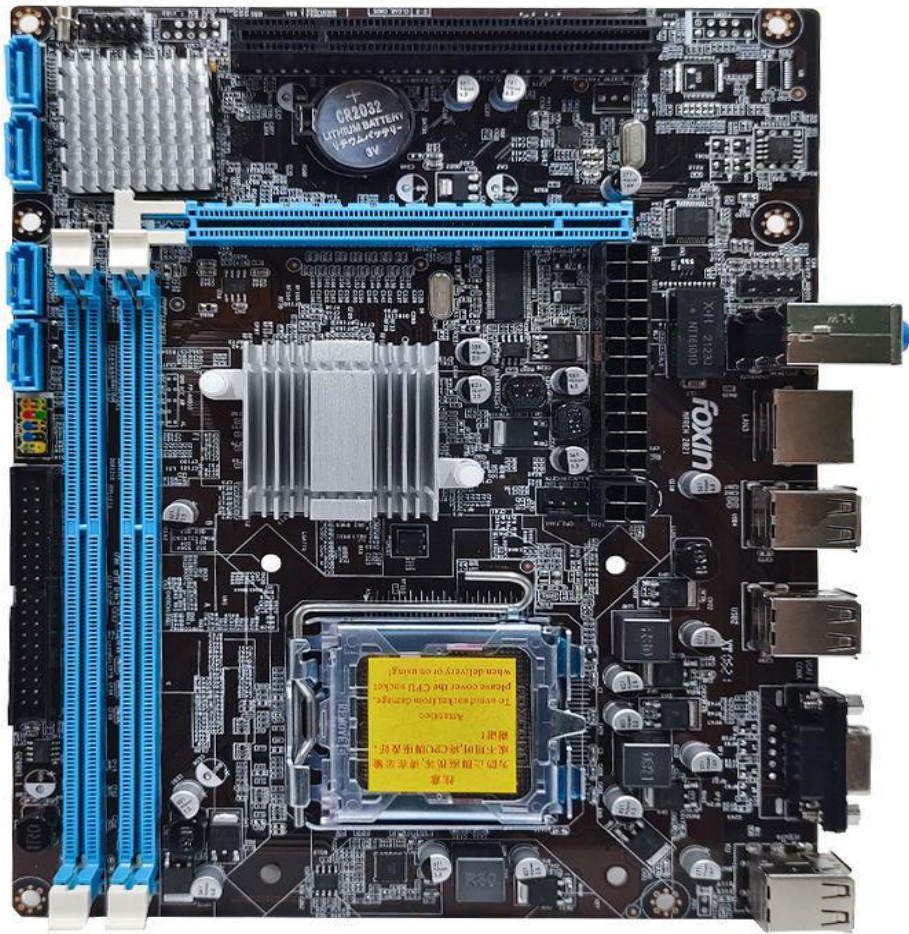
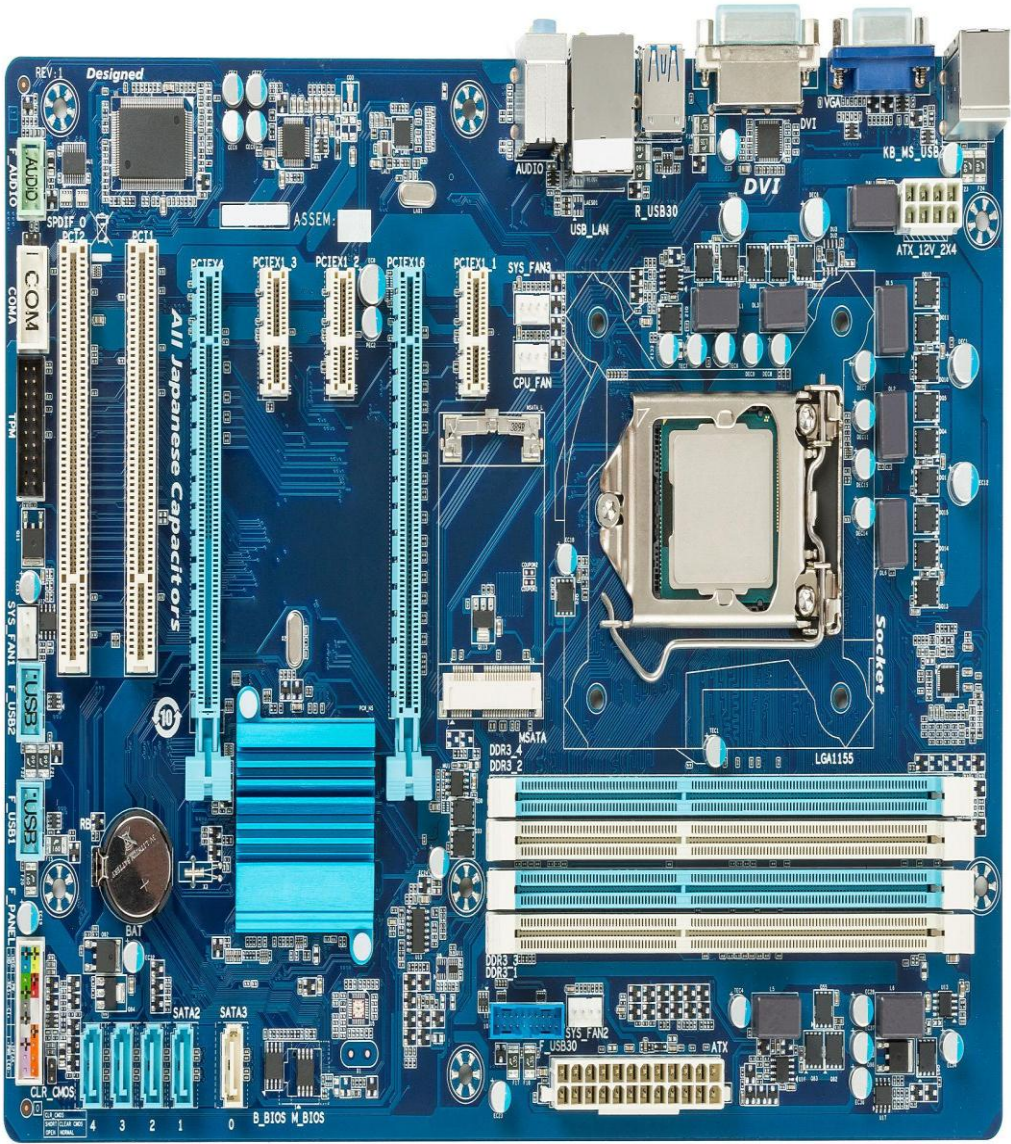
2. Chipset:

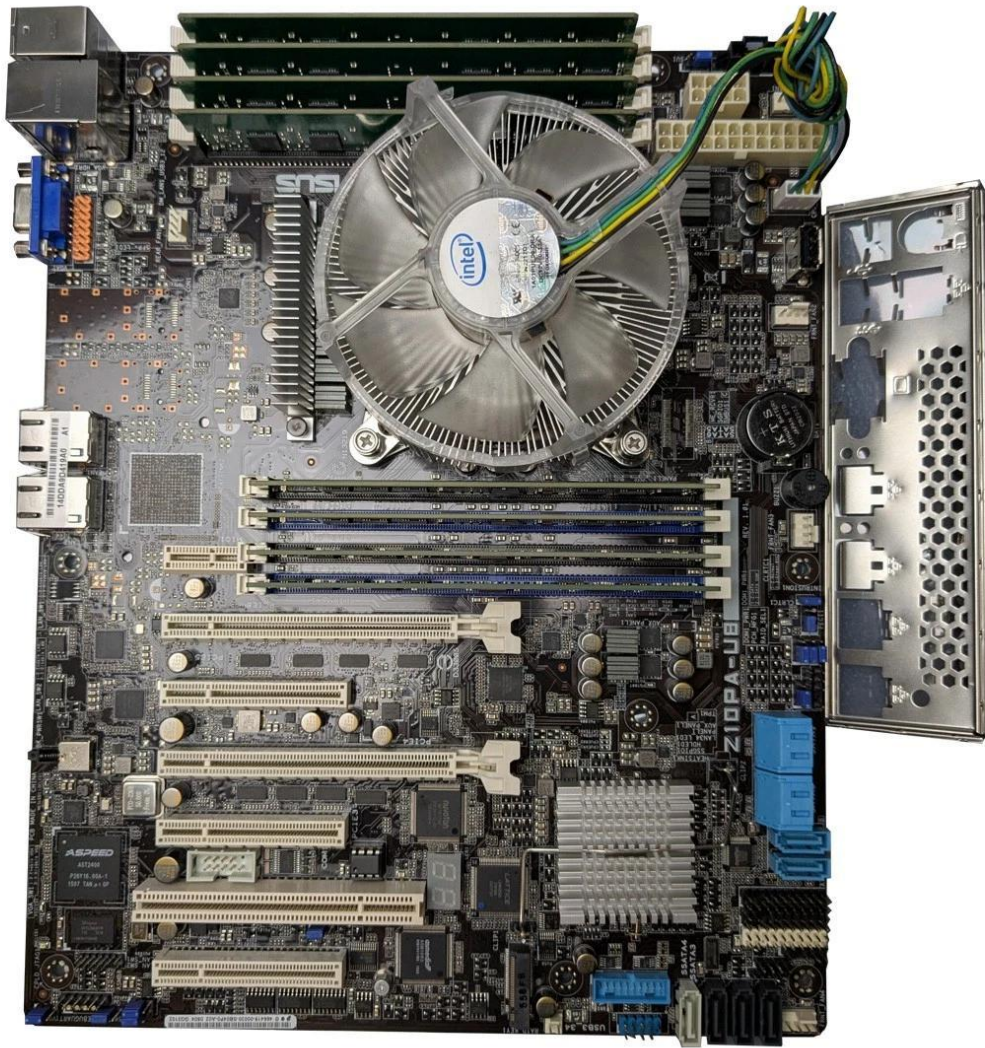
- The chipset is a group of electronic components on the motherboard that manages communication between the CPU and other hardware devices.

Q4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

ANS.







❖ **Label these 5 motherboard ports/slots**

1. **CPU Socket:** The square socket in the center where the processor (CPU) is installed.
 2. **RAM (DIMM) Slots:** The long vertical slots beside the CPU used to install memory modules.
 3. **PCIe x16 Slot:** The long expansion slot used to install a graphics card (GPU).
 4. **SATA Ports:** Small L-shaped connectors (usually grouped together) used to connect HDDs, SSDs, or DVD drives.
 5. **24-Pin ATX Power Connector:** The large connector on the edge of the motherboard that supplies power from the power supply (PSU).
-